AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

APRIL/ MAY 2017

Physics Paper II

Time: 2 hours 25 minutes Marks: 40

AKU May Examination 2017

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer ALL questions. Questions 9, 10 & 11 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 4 Marks)

A pendulum of length 1.00 m has a period of 2.01 s, when it is allowed to swing at a height of 8954.0 m on Mount Everest.

i. Calculate the gravitational acceleration of the pendulum. (2 Marks)

ii. Explain the difference, if the same pendulum is allowed to swing at sea level. (1 Mark)

iii. Give a scientific reason for the difference noted in part (ii). (1 Mark)

Q.2. (Total 2 Marks)

i. Define power of a lens. (1 Mark)

ii. Write its S.I. unit. (1 Mark)

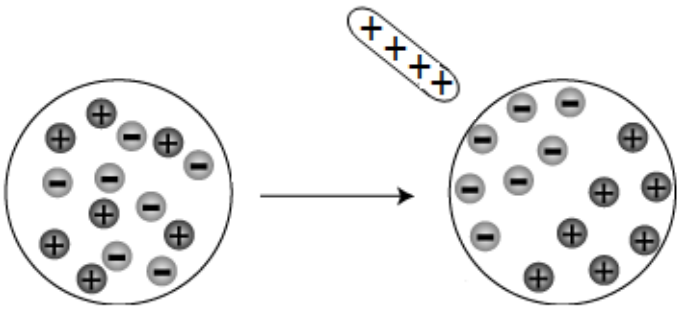
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Q.3. (Total 3 Marks)

If the real depth of a swimming pool is 5.50 m, then calculate the apparent depth of the swimming pool.
(**Note**: The refractive index *n* of water is 1.33.)

Q.4. (Total 3 Marks)

i. Identify the given phenomenon in the given diagram. (1 Mark)

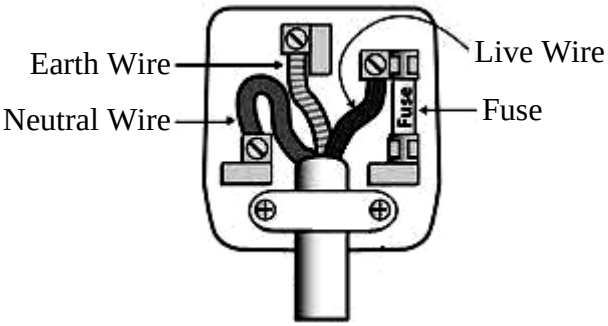


ii. Mention any TWO hazards of the phenomenon identified in part (i). (2 Marks)

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Q.5. (Total 3 Marks)

The given figure shows a power plug.



i. How is the safety of an electrical appliance secured by using a fuse in the power plug? (1 Mark)

_____

_____

ii. If the rating of the fuse shown in the given figure is 12 ampere and it is working at 240 volt, then calculate the power of the fuse. (2 Marks)

_____

_____

_____

_____

Q.6. (Total 3 Marks)

Write THREE uses of transformers in our electric power system.

_____

_____

_____

_____

_____

_____

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Q.7. (Total 3 Marks)

A truth table of a logic gate is given below.

A	B	$X = \overline{A.B}$
0	0	1
1	0	1
0	1	1
1	1	0

i. Identify the given logic gate. (1 Mark)

_____

ii. Draw the symbol of the logic gate. (1 Mark)

iii. Write its mathematical operation. (1 Mark)

_____

Q.8. (Total 4 Marks)

i. Complete the given statement: (1 Mark)

Alpha (α) decay is the emission of helium _____.

ii. Determine whether the given decay process is possible or not and show your working.

$^{214}_{84}\text{Po} \rightarrow {}^{208}_{82}\text{Pb} + {}^{4}_{2}\alpha$ (3 Marks)

_____

_____

_____

_____

_____

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Q.9. (Total 5 Marks)

EITHER

a.

i. Write formula for the speed of a wave. (1 Mark)

ii. 'Waves are only the means of energy transfer, without the transfer of matter.' (4 Marks)

Describe any FOUR examples in the light of the above statement.

OR

b. Zain is a social worker who wants to create awareness regarding the use of acoustic protection in the major cities of Pakistan. He is planning to publish a brochure to explain the advantages of acoustic protection to the general public.

As a physics student, suggest FIVE advantages of acoustic protection which should be published in this brochure. (5 Marks)

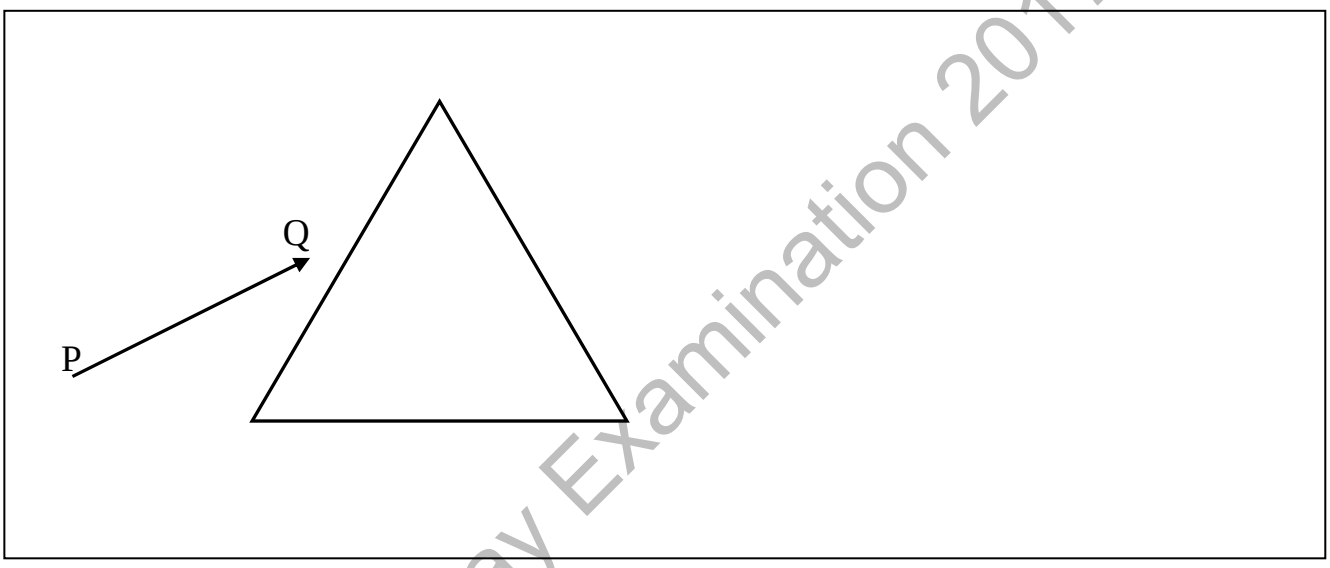
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Q.10. (Total 5 Marks)

EITHER

a. The given diagram shows an incident ray PQ is reaching a glass prism. Complete the ray diagram illustrating the following.

i. Normal (1 Mark)
ii. Refracted ray (1 Mark)
iii. Emergent ray (1 Mark)
iv. Angle of deviation (1 Mark)
v. Angle of emergent (1 Mark)

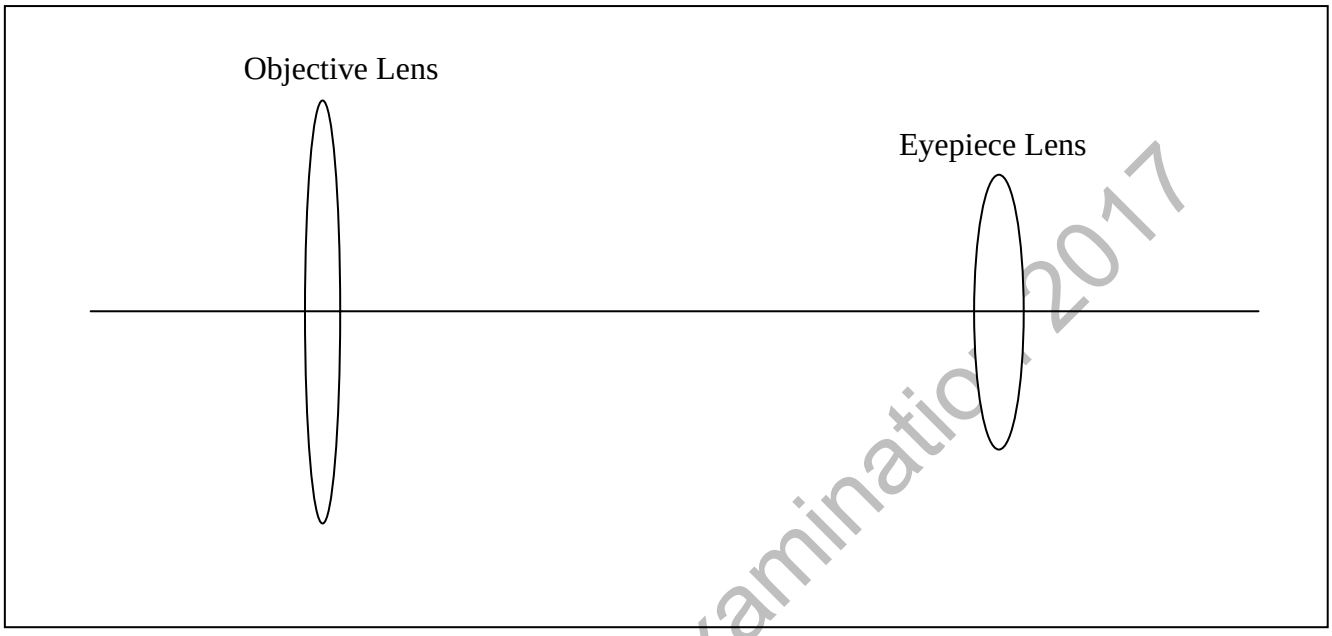


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OR

b. An astronomer uses a telescope to observe stars at night.

Complete the following ray diagram to show the refraction of light which allows him to see the stars through the telescope. (5 Marks)

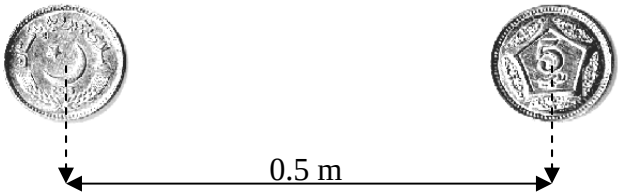


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Q.11. (Total 5 Marks)

EITHER

a. The given two coins carry identical charge and experience a repulsive force of 5 N.



Calculate the value of charge on each coin. (5 Marks)

(**Note**: The value of the constant of proportionality 'k' is 9×10^9 Nm^2/C^2.)

OR

b. Assume a charged particle of 80 C is to be transferred through a potential difference of 40 V in time 20 s. Calculate the following:

i. Work done (3 Marks)

ii. Power (2 Marks)

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